

CBSE Test Paper 01
Chapter 6 Work Energy & Power

1. A 0.800-kg ball is tied to the end of a string 1.60 m long and swung in a vertical circle. Calculate the total work done on the ball by (i) the tension in the string and (ii) gravity for motion along the semicircle from the lowest to the highest point on the path. **1**
 - a. 0, -281 J
 - b. 0, -251 J
 - c. 0, -2.51 J
 - d. 0, -25.1 J
 2. A bullet of mass 0.012 kg and horizontal speed 70 m s^{-1} strikes a block of wood of mass 0.4 kg and instantly comes to rest with respect to the block. The block is suspended from the ceiling by means of thin wires. Calculate the height to which the block rises. Also, estimate the amount of heat produced in the block. **1**
 - a. 72.2 cm, 28.5 J
 - b. 21.2 cm, 28.5 J
 - c. 29.2 cm, 29.5 J
 - d. 23.2 cm, 38.5 J
 3. A rain drop of mass 2.00 g falls from a height 1.00 km. It hits the ground with a speed of 50.0 m/s. What is the work done by the resistive forces if any? (Take $g = 10 \text{ m s}^{-2}$) **1**
 - a. - 27.5 J
 - b. - 22.5 J
 - c. - 19.5 J
 - d. - 17.5 J
 4. A 0.800-kg ball is tied to the end of a string 1.60 m long and swung in a vertical circle. (a) During one complete circle, starting anywhere, calculate the total work done on the ball by (i) the tension in the string and (ii) gravity **1**
 - a. 25 J, -25 J
 - b. 10 J, 10 J
 - c. 30 J, 30 J
 - d. 0, 0 J
 5. A block having a mass of 0.80 kg is given an initial velocity 1.2 m/s to the right and
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collides with a spring of negligible mass and force constant $k = 50 \text{ N/m}$. Assuming the surface to be frictionless, calculate the maximum compression of the spring after the collision. **1**

- a. 0.15 m
- b. 0.20 m
- c. 0.10 m
- d. 0.25 m

6. Does the amount of work done depend upon the fact that how fast is a load raised or moved in the direction of force? **1**
 7. Write the characteristics of displacement? **1**
 8. Is a collision between two particles possible even without any physical contact between them? **1**
 9. What is the work done by the centripetal force? Explain. **2**
 10. If one mole of a monatomic gas is mixed with 3 moles of a diatomic gas. What is the molecular specific heat of the mixture at constant volume? **2**
 11. If the volume of block of metal changes by 0.12% when it is heated through 200°C . What is the co-efficient of linear expansion of the metal? **2**
 12. Prove that in an elastic collision in one dimension the relative velocity of approach before impact is equal to the relative velocity of separation after impact. **3**
 13. State work-kinetic energy theorem and prove it analytically. **3**
 14. The nucleus Fe^{57} emits a γ -ray of energy 14.4 keV. If the mass of the nucleus is 56.935 amu, calculate the recoil energy of the nucleus. [Take, $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$] **3**
 15. Prove that when a particle suffers an oblique elastic collision with another particle of equal mass and initially at rest, the two particles would move in mutually perpendicular directions after collisions. **5**
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